

假设 我们有 a biased coin, 但是 we don't know what the bias is. To estimate the bias, we toss the coin

$n$  times 和 count how many Heads we observe. 那么 our estimate of the bias is 给定 by  $\hat{p} = \frac{1}{n} S$ , 其中

$n$

$S$  is the number of Heads in the  $n$  tosses. Is this a good estimate? 令  $p$  表示 the 真 bias of the coin,

$n$

其中  $p$  is unknown to us. 因为  $E[S] = np$ , we see that the estimator  $\hat{p}$  always has the correct expected

$n$

value:  $E[\hat{p}] = \frac{1}{n} E[S] = p$ . What is the role of  $n$  here?

Why do we take the average of many samples, rather

$n$

than just one? As we shall see, as  $n$  gets larger, the estimate  $\hat{p}$  gets closer to the desired value  $p$ . This is a

manifestation of the Law of Large Numbers, which corresponds to our own empirical experience and which we shall prove formally at the end of this note.

How large should  $n$  be to guarantee that our estimate

$\hat{p}$  is within an error  $\epsilon$  of the true bias  $p$ , i.e.,  $|\hat{p} - p| \leq \epsilon$ ?

The 答案 is that we can never guarantee with absolute certainty that  $|\hat{p} - p| \leq \epsilon$ . This is because  $S$  is

$n$

a random 变量 that can take any 整数 values between 0 和  $n$ , 和 因此  $\hat{p} = \frac{1}{n} S$  is also a random

$n$

variable that can take any values between 0 and 1.

So regardless of the value of the true bias  $p \in (0, 1)$ , it is

possible that in our experiment of  $n$  coin tosses we observe  $n$  Heads (this happens with 概率  $p^n$ ), in

which case  $S = n$  and  $\hat{p} = 1$ .

Similarly, it is also possible that all  $n$  coin tosses come up Tails (this happens

$n$

with probability  $(1 - p)^n$ ), in which case  $S = 0$  and  $\hat{p} = 0$ .

$n$

So instead of requiring that  $|\hat{p} - p| \leq \epsilon$

with absolute certainty, we relax our requirement and only demand

that  $|\hat{p} - p| \leq \epsilon$  with confidence  $1 - \delta$ , 即,  $P[|\hat{p} - p| \leq \epsilon] \geq 1 - \delta$ .

This means there is a small probability

$\delta$

that we make an error of more than  $\epsilon$ , but with high probability (at least  $1 - \delta$ ) our estimate is very close to  $p$ . Now we can state our result: to guarantee that

$P[|\hat{p} - p| \leq \epsilon] \geq 1 - \delta$ , it suffices to toss the coin  $n$  times where

1

$n \geq \frac{1}{4\epsilon^2\delta}$ .

$4\epsilon^2\delta$

To prove such a result, we use a mathematical tool called Chebyshev's Inequality, which provides a quanti-

tative, probabilistic bound on how far away a random variable can be from its expected value.

1 马尔可夫's Inequality

Before discussing 切比雪夫's inequality, we first 证明 the following simpler 界, 其中 applies only

nonnegative random variables (i.e., r.v.'s which take only values  $\geq 0$ ).  
 Theorem 19.1 (马尔可夫's Inequality).  
 For a nonnegative random variable  $X$  (i.e.,  $X(\omega) \geq 0$  for all  $\omega \in \Omega$ )  
 with finite mean,  
 $E[X]$   
 $P[X \geq c] \leq \frac{E[X]}{c}$ ,  
 for any positive constant  $c$ .  
 CS70, Spring 2026, Note 19.1

证明. Let  $A$  denote the range of  $X$  and consider any constant  $c \in A$ . 那么,  
 $E[X] = \sum_{a \in A} a \times P[X = a]$   
 $\geq \sum_{a \geq c} a \times P[X = a]$   
 $\geq \sum_{a \geq c} c \times P[X = a]$   
 $= c \sum_{a \geq c} P[X = a]$   
 $= c P[X \geq c]$ ,

where the first line is just the definition of expectation, while the second line follows from the fact that  $X$  is a nonnegative random variable (i.e.,  $a \geq 0$  for all  $a \in A$ ).  
 Rearranging the last inequality gives us the desired result.  
 (笔记: Setting  $c = 2E[X]$  in 马尔可夫's inequality, the statement is that at most 1/2 the people can be at least twice the average.)

As a slicker proof can be provided using the indicator function  $I\{\cdot\}$ , defined for any event  $E$  as  
 $I\{E\} = \begin{cases} 1, & \text{if } E \text{ is true,} \\ 0, & \text{if } E \text{ is false.} \end{cases}$

Alternative proof of Theorem 19.1. 因为  $X$  is a nonnegative random variable 和  $c > 0$ , 我们有 对所有

$\omega \in \Omega$ ,  
 $X(\omega) \geq c I\{X(\omega) \geq c\}$ , (1)  
 since the right hand side is 0 if  $X(\omega) < c$  and is  $c$  if  $X(\omega) \geq c$ .

Multiplying both sides by  $P[\omega]$  and summing over  $\omega \in \Omega$  gives

$E[X] \geq c E[I\{X \geq c\}] = c P[X \geq c]$ ,

其中 the first inequality follows from (1) 和 the last equality from the fact that  $I\{X \geq c\}$  is an indicator random variable.

If we have a random variable  $Y$  that is not necessarily nonnegative, then the same line of argument as above

can be applied to prove the following result for the absolute value  $|Y|$  of  $Y$ :

定理 19.2 (Generalized 马尔可夫's Inequality). 令  $Y$  be an arbitrary random variable with 有限的 mean. 那么, for any positive constants  $c$  and  $r$ ,

$E[|Y|^r]$   
 $P[|Y| \geq c] \leq \frac{E[|Y|^r]}{c^r}$ .

证明. Since  $c > 0$  and  $r > 0$ , we have

$|Y|r \geq |Y|rI\{|Y| \geq c\} \geq crI\{|Y| \geq c\}.$   
 (Notethatthelastinequalitywouldnotholdifrwere negative.)  
 Takingtheexpectationofbothsidesgives  
 $E[|Y|r] \geq crE[I\{|Y| \geq c\}] = crP[|Y| \geq c],$   
 anddividingbycr leadstothedesiredresult.  
 CS70, Spring2026, Note19 2

例子: CoinTosses

Consider tossing a fair coin  $n$  times and let  $X$   
 denote the number of heads observed. What is the probability  
 of observing more than  $3n$  heads?  
 令' apply Markov' sinequality to obtain an upper bound on  $P[X \geq 3n]$ .

4 4  
 Since  $X \sim \text{Bin}(n, 1/2)$ , we have  $E[X] = n/2$ , 所以  
 2 2  
 $E[X] = n/2$   
 $P[X \geq 3n/2] \leq \frac{E[X]}{3n/2} = \frac{n/2}{3n/2} = \frac{1}{3},$   
 4 3n 3  
 4  
 which does not depend on  $n$ . Is this a good upper bound?  
 The exact answer for  $P[X \geq 3n/2]$  is given by  
 4  
 $P[X \geq 3n/2] = \sum_{k=3n/2}^n \binom{n}{k} 2^{-n},$   
 4 k 2n  
 (cid:108) 3 (cid:109)  
 $k = n$   
 4  
 which is  $\approx 5.5 \times 10^{-2}$  for  $n=10$  and  $\approx 2.8 \times 10^{-7}$  for  $n=100$ .  
 As these numerical values show,  $P[X \geq 3n/2]$   
 4

is a decreasing function of  $n$ , and so Markov' sinequality does not provide a very good upper bound for this  
 例子. In the next section, we will see how to obtain an improved bound.

2 切比雪夫' s Inequality

我们有 seen that, intuitively, the 方差 (或, more correctly the  
 标准差) is a measure of  
 "spread," or deviation from the mean.

We can now make this intuition quantitatively precise:

Theorem 19.3 (切比雪夫' s Inequality). For a random variable  $X$   
 with finite expectation  $E[X] = \mu$ ,

$\text{Var}(X)$   
 $P[|X - \mu| \geq c] \leq \frac{\text{Var}(X)}{c^2}, \quad (2)$

c2  
 for any positive constant  $c$ .

证明. Define  $Y = (X - \mu)^2$  and note that  $E[Y] = \text{Var}(X)$ .

Also, notice that the event that we  
 are interested in,  $|X - \mu| \geq c$ , is exactly the same as the  
 event  $Y \geq c^2$ . 因此,  $P[|X - \mu| \geq c] = P[Y \geq c^2]$ . 此外,  $Y$

is obviously nonnegative, so we can apply Markov' sinequality in Theorem 19.1  
 to get

$E[Y] = \text{Var}(X)$   
 $P[|X - \mu| \geq c] = P[Y \geq c^2] \leq \frac{E[Y]}{c^2} = \frac{\text{Var}(X)}{c^2}.$   
 c2 c2

This completes the proof.

Here is a simpler proof using Generalized Markov' sinequality from Theorem 19.2:

Alternative proof of Theorem 19.3. 定义  $Y = |X - \mu|^2$ . 那么, 因为  $|Y| = Y$   
 $= |X - \mu|^2 = (X - \mu)^2$ , 我们有

(cid:104) (cid:105)

$E|Y|^2 = E[(X - \mu)^2] = \text{Var}(X)$ . 因此, (2) follows from applying 定理 19.2 to  $Y = X - \mu$  with  $r=2$ .

令's pause to 考虑 what 切比雪夫's inequality says. It tells us that the 概率 of any 给定

deviation,  $c$ , from the mean, either above it 或 below it (笔记 the absolute value sign), is at most

$\frac{\text{Var}(X)}{c^2}$ .

$\frac{1}{c^2}$

As expected, this deviation 概率 will be small 如果 the 方差 is small. An immediate 推论 of

切比雪夫's inequality is the following:

CS70, Spring 2026, Note 19.3

Corollary 19.1. For any random variable  $X$

with finite expectation  $E[X] = \mu$  and finite standard deviation

(cid:112)

$\sigma = \sqrt{\text{Var}(X)}$ ,

1

$P[|X - \mu| \geq k\sigma] \leq \frac{1}{k^2}$ ,

$k^2$

for any constant  $k > 0$ .

证明. Plug  $c = k\sigma$  into Chebyshev's inequality.

所以, 对于 例子, we see that the 概率 of deviating from the mean by more than (say) two standard

deviations on either side is at most 1. In this sense, the

标准差 is a good working 定义 of

4

the "width" 或 "spread" of a distribution.

In some special cases, it is possible to get tighter bounds on the 概率 of deviations from the mean.

然而, 对于 general random variables that only have means 和 variances, 切比雪夫's inequality is es-

entially the only tool. Its power derives from the fact

that it can be applied to any random 变量 with a

mean and a variance.

例子: Coin Tosses Revisited

令's revisit the coin toss 例子 considered above 和 apply 切比雪夫's inequality to obtain an upper

bound on  $P[X \geq 3n]$ . Recalling  $E[X] = n$ , we obtain

$\frac{4}{n^2}$

$P[X \geq 3n] = P[X - n \geq 2n] \leq P[(X - n)^2 \geq 4n^2]$

$\leq \frac{\text{Var}(X)}{4n^2} = \frac{n}{4n^2} = \frac{1}{4n}$ .

$\frac{1}{4n}$

4

Since  $X$  is binomial with parameters  $(n, 1/2)$ , we have  $\text{Var}(X) = n(1/2)(1 - 1/2) = n/4$ , so we obtain

$\frac{1}{4n}$

$\text{Var}(X) = \frac{n}{4}$

$P[X \geq 3n] \leq \frac{1}{4n}$ ,

$\frac{1}{4n}$

4

which is much better than the constant bound of  $\frac{1}{4}$

given by Markov's inequality.

3

3 Applications

In this section, we discuss applications of Chebyshev's inequality to estimation problems.

3.1 Estimating the Bias of a Coin

令 us go back to our motivating 例子 of estimating the bias of a coin. 回忆 我们有 a coin of unknown bias  $p$ , and our estimate of  $p$  is  $\hat{p} = 1S/n$  where  $S$  is the number of Heads in  $n$  coin tosses.

$n$   $n$   $n$   
 As usual, we will find it helpful to write  $S = X_1 + \dots + X_n$ , where  $X_i = 1$  if the  $i$ -th coin toss comes up Heads and  $X_i = 0$  otherwise, and the random variables  $X_1, \dots, X_n$  are independent and identically distributed. 那么  
 $i = 1$  to  $n$   
 $E[X_i] = P[X_i = 1] = p$ , so by linearity of expectation,  
 $i = 1$  to  $i$   
 $E[\hat{p}] = E(1/n \sum_{i=1}^n X_i) = 1/n \sum_{i=1}^n E[X_i] = p$ .  
 $n$   $n$   $n$   $i$   
 $i=1$

CS70, Spring 2026, Note 19 4

What about the variance of  $\hat{p}$ ? Note that since the  $X_i$

$i$   
 's are independent, the variance of  $S$

$n$   
 $= \sum_{i=1}^n \text{Var}(X_i)$   
 $X_i$   
 $i$   
 are equal  
 to the sum of the variances:

$1/n^2 \sum_{i=1}^n \sigma^2$   
 $\text{Var}(\hat{p}) = \text{Var}(1/n \sum_{i=1}^n X_i) = 1/n^2 \sum_{i=1}^n \text{Var}(X_i) = \sigma^2/n$ ,  
 $n$   $n$   $n^2$   $n$   $n^2$   $i$   $n$   
 $i=1$

where we have written  $\sigma^2$  for the variance of each of the  $X_i$ .

$i$   
 所以 we see that the variance of  $\hat{p}$  is inversely proportional to  $n$ .  
 This fact ensures that, as we take larger and

larger sample sizes  $n$ , the probability that we deviate much from the expectation goes smaller and smaller.

令's now use 切比雪夫's inequality to 图 out how large  $n$  has to be to ensure a specified accuracy in our estimate of the true bias  $p$ .

As we discussed in the beginning of this note, a natural way to measure this is for us to specify two parameters,  $\epsilon$  and  $\delta$ , both in the range  $(0, 1)$ .

The parameter  $\epsilon$  controls the error we are prepared to tolerate in our estimate, and  $\delta$  controls the confidence we want to have in our estimate.

Applying Chebyshev's inequality, we have

$\text{Var}(\hat{p}) = \sigma^2/n$   
 $P[|\hat{p} - p| \geq \epsilon] \leq \frac{\sigma^2/n}{\epsilon^2} = \frac{\sigma^2}{n\epsilon^2}$ .

To make the right hand side less than the desired value  $\delta$ , we need to set  $\sigma^2/n \leq \delta$ .

$n \geq \frac{\sigma^2}{\epsilon^2 \delta}$ . (3)

Now 回忆  $\sigma^2 = \text{Var}(X)$  is the 方差 of a single sample  $X$ . 所以, 因为  $X$  is an indicator random

variable with  $P[X = 1] = p$ , we have  $\sigma^2 = p(1-p)$ , and the inequality from (3) becomes  $n \geq \frac{p(1-p)}{\epsilon^2 \delta}$ . 因为

$i \in \{1, 2, \dots, n\}$   
 $p(1-p)$  is maximized when  $p = 1/2$ , we can conclude that it is sufficient to choose  $n$  such that:  
 $n \geq \frac{1}{4\epsilon^2\delta}$  (4)  
 $p \in [1/2 - \delta, 1/2 + \delta]$   
 as we claimed earlier.  
 For example, plugging in  $\epsilon = 0.1$  and  $\delta = 0.05$ , we see that a sample size of  $n = 500$  is sufficient to get an estimate  $\hat{p}$  that is accurate to within an error of  $0.1$  with probability at least  $95\%$ .  
 As a concrete example, consider the problem of estimating the proportion of Democrats in the US population, by taking a small random sample. We can model this as the problem of estimating the bias of a coin

above, where each coin toss corresponds to a person that we select randomly from the entire population, and the coin tosses are independent<sup>2</sup>. Our calculation above shows that to get an estimate  $\hat{p}$  即 accurate to

within an error of  $0.01$  with probability at least  $95\%$ , it suffices to sample  $n = 50,000$  people<sup>3</sup>. In particular, notice that the size of the sample is 独立的 of the total size of the population! This is how polls can

accurately estimate quantities of interest for a population of several hundred million while sampling only a fairly small number of people.  
<sup>1</sup> Use calculus to prove this.

<sup>2</sup> We are assuming here that the sampling is done “with replacement”; i.e., we select each person in the sample from the entire

population, including those we have already picked. So there is a small chance that we will pick the same person twice.

<sup>3</sup> Actually the sample size needed is quite a bit smaller than this, as you will see in later classes such as CS174 using more

sophisticated concentration bounds. But this is certainly a valid upper bound.  
 CS70, Spring 2026, Note 19.5

### 3.2 Estimating a General 期望

What 如果 we wanted to estimate something a little more 复数 than the bias of a coin? 对于 例子, 假设 we want to estimate the average wealth of people in the US. We can model this as the 题目 of estimating the expected value of an unknown 概率 分布. 那么 we can use exactly the same scheme as before, except that now we sample the random variables  $X_1, X_2, \dots, X_n$  independently from our unknown distribution.

Clearly  $E[X] = \mu$ , the expected value that we are trying to estimate. Our estimate of

$\mu$  will be  $\hat{\mu} = \frac{1}{n} \sum_{i=1}^n X_i$

$i$   
 , for a suitably chosen sample size  $n$ .  
 Following the same calculation as before, 我们有  $E[\hat{\mu}] = \mu$  和  
 $\text{Var}(\hat{\mu}) = \frac{\sigma^2}{n}$ , 其中  $\sigma^2 = \text{Var}(X)$  is the  
 $n$   
 variance of the  $X$ . (Recall that the only facts we used about the  $X$   
 was that they were independent and had  
 $i$   
 the same 分布 — actually the same 期望 和 方差 would be enough:  
 why?) This time,

然而, since we do not have any a priori bound on the mean  $\mu$ , it makes more sense to let  $\epsilon$   
 be the relative

error, i. e., we wish to find an estimate  $\hat{\mu}$   
 that is within an additive error of  $\epsilon \mu$ :

$$P[|\hat{\mu} - \mu| \geq \epsilon \mu] \leq \delta.$$

Using (3), but substituting  $\epsilon \mu$

in place of  $\epsilon$ , it is enough for the sample size to satisfy

$$n \geq \frac{1}{\epsilon^2 \delta} \quad (5)$$

Here  $\epsilon$  和  $1 - \delta$  are the desired relative error 和 confidence  
 level respectively. Now of course we don't

know the other two quantities,  $\mu$  and  $\sigma^2$ , appearing in (5).

In practice, we would use a lower bound on  $\mu$  和  
 an upper bound on  $\sigma^2$  (just as we used an upper bound on

$p(1-p)$  in the coin tossing problem). Plugging

these bounds into (5) will ensure that our sample size is large enough.  
 对于 例子, in the average wealth 题目 we could probably safely

take  $\mu$  to be at least (say) \$20,000  
 (probably more). 然而, the existence of very wealthy people

such as Bill Gates means that we would  
 need to take a very high value 对于 the 方差  $\sigma^2$ . Indeed, 如果 存在

at least one individual with wealth  
 \$50 billion in a population of size 325 million, 那么 assuming

a relatively small value of  $\mu$  means that the  
 方差 must be at least about  $(50 \times 10^9)^2 \approx 7.7 \times 10^{12}$ . 存在 really no

way around this 题目 with

$325 \times 10^6$   
 simple uniform sampling:

the uneven distribution of wealth means that the variance is inherently very large,

and we will need a huge number of samples before we are likely to find anybody who is immensely wealthy.

But if we don't include such people in our sample, then our estimate will be way too low.

#### 4 The Law of Large Numbers

The estimation method we used in the previous sections is  
 based on a principle that we accept as part of

everyday life: 即, the Law of Large Numbers (LLN). This  
 asserts that, 如果 we observe some random

变量 many times, 和 take the average of the observations, 那么  
 this average will 收敛 to a single

value, which is of course the expectation of the random variable.

In other words, averaging tends to smooth

out any large fluctuations, and the more averaging we do the better the smoothing.

Theorem 19.4 (Law of Large Numbers). Let  $X_1, X_2, \dots$ , be a sequence of i. i. d. (independent and identically

distributed) random variables. Then

$X_1, X_2, \dots, X_n$  distributed) random variables with common 有限的 期望  $E[X] = \mu$  对所有  $i$ . 那么, their partial sums  $S_n = X_1 + X_2 + \dots + X_n$  satisfy  

$$P(|S_n - n\mu| \geq \epsilon n) \rightarrow 0 \text{ as } n \rightarrow \infty,$$
 for every  $\epsilon > 0$ , however small.  
 CS70, Spring 2026, Note 19.6

证明. 令  $\text{Var}(X) = \sigma^2$  be the common 方差 of the r.v.'s; we 假设 that  $\sigma^2$  is 有限的. 4 With this

(relatively mild) assumption, the LLN is an immediate consequence of 定理 19.3. 因为  $X_1, X_2, \dots$  are i.i.d. random variables with  $E[X] = \mu$  和  $\text{Var}(X) = \sigma^2$ , 我们有  $E(S_n) = n\mu$  和  $\text{Var}(S_n) = n\sigma^2$ , 所以 by 切比雪夫' sinequality we have  

$$P(|S_n - n\mu| \geq \epsilon n) \leq \frac{\text{Var}(S_n)}{\epsilon^2 n} = \frac{n\sigma^2}{\epsilon^2 n} \rightarrow 0 \text{ as } n \rightarrow \infty.$$
 Notice that the LLN says that the probability of any deviation  $\epsilon$  from the mean, however small, tends to zero as the number of observations  $n$  in our average tends to infinity. 因此, by taking  $n$  large enough, we can make the probability of any given deviation as small as we like.

4 If  $\sigma^2$  is not finite, the LLN still holds but the proof requires more advanced mathematical tools.  
 CS70, Spring 2026, Note 19.7